

Modelling Electron Polarons in Rutile TiO_2 using DFT+U

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1 Introduction

Rutile TiO_2 is a transition metal oxide where excess electrons can localize on Ti sites and form small electron polarons. Standard PBE often underestimates the band gap and can over-delocalize Ti $3d$ electrons. Therefore, DFT+U was used to improve the description of Ti d states and to study electron localization.

In this work, bulk rutile TiO_2 , a neutral $3 \times 3 \times 4$ supercell, and localized polaron configurations were calculated. Two methods were used to add one excess electron: manually increasing `NELECT`, and substituting one Ti atom by Nb.

2 Computational setup

All calculations were performed using VASP with PBE PAW potentials. For DFT+U calculations, the Dudarev approach was used with

$$U_{\text{eff}} = U - J = 3.9 \text{ eV}$$

applied to Ti d states. In the VASP `INCAR`, this corresponds to

$$\text{LDAUL} = 2 \text{ } -1$$

for the species order Ti O, where 2 applies U to Ti d orbitals and -1 means no U correction for O.

3 Standard DFT PBE calculation for rutile TiO_2

The rutile TiO_2 primitive cell was prepared from the ICSD #63710 structure. The primitive cell contains 2 Ti and 4 O atoms. The POTCAR was constructed using the PBE PAW potentials in the same order as the POSCAR, namely Ti followed by O_s.

First, the internal atomic positions were relaxed using PBE. The relaxed `CONTCAR` was then used as the input structure for the fixed electronic structure calculations.

Using the relaxed structure, a fixed PBE self-consistent calculation was performed. The eigenvalues from the SCF/DOS run show that the highest occupied state is at -1.741759 eV and the lowest unoccupied state is at 0.000009 eV relative to the Fermi level. This gives a PBE SCF eigenvalue gap of

$$E_g^{\text{PBE}} = 1.741768 \text{ eV}.$$

The DOS plot shows a clear gap between the valence and conduction states. A separate band structure calculation along a high symmetry k-point path gives a similar gap of about 1.680138 eV. The small difference is expected because the SCF/DOS calculation uses a regular k-point mesh, while the band structure uses a high symmetry k-point path.

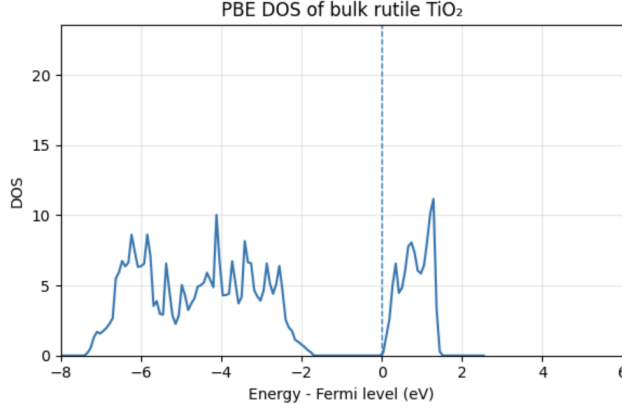


Figure 1: PBE DOS of bulk rutile TiO_2 .

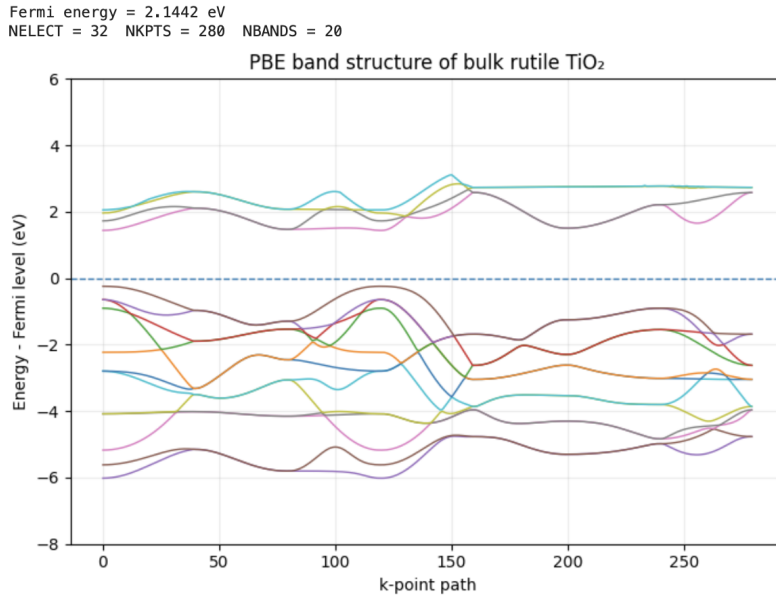


Figure 2: PBE band structure of bulk rutile TiO_2 .

4 DFT+U calculation for rutile TiO_2

The fixed structure electronic calculation was repeated using DFT+U with $U_{\text{eff}} = 3.9$ eV applied to Ti d states. The relaxed PBE structure was reused, and the main change was in the INCAR, where the DFT+U correction was included.

From the DFT+U SCF/DOS eigenvalues, the highest occupied state was found at -2.260820 eV and the lowest unoccupied state at 0.000019 eV relative to the Fermi level. This gives a DFT+U gap of

$$E_g^{\text{DFT+U}} = 2.260839 \text{ eV}.$$

The DFT+U DOS and band structure show a larger band gap than standard PBE. This is expected because the U correction shifts the Ti d conduction states and improves the description of localized Ti d electrons.

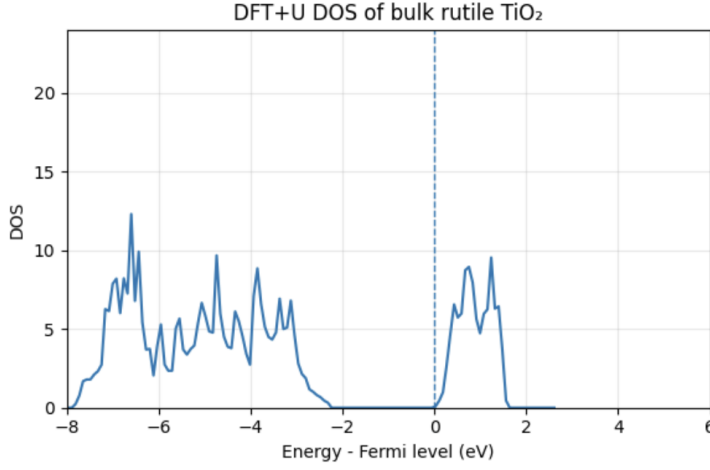


Figure 3: DFT+U DOS of bulk rutile TiO_2 .

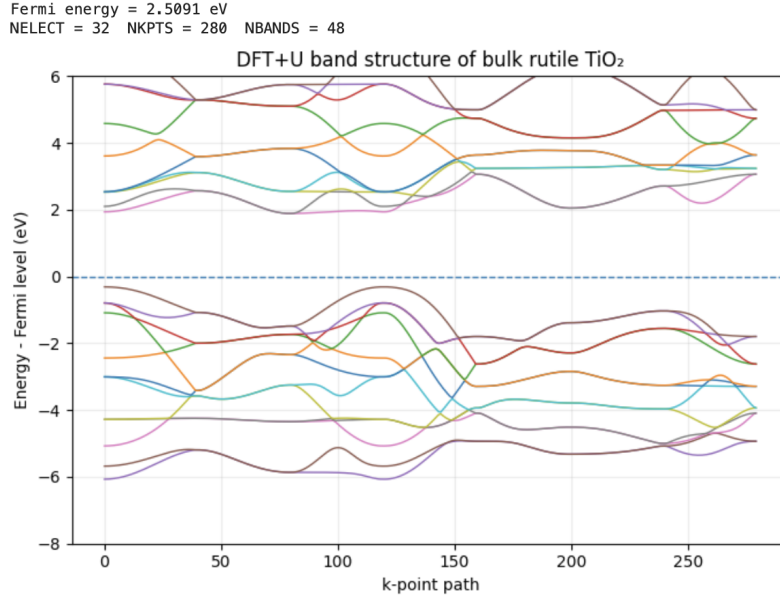


Figure 4: DFT+U band structure of bulk rutile TiO_2 .

Table 1: Comparison of PBE and DFT+U band gaps for bulk rutile TiO_2 .

Calculation	SCF/DOS gap (eV)	Band structure gap (eV)
PBE	1.741768	1.680138
DFT+U	2.260839	2.200752

5 DFT+U calculation using a $3 \times 3 \times 4$ supercell

A $3 \times 3 \times 4$ rutile TiO_2 supercell was generated from the primitive cell using a Python script instead of VESTA or p4vasp. The script repeated the primitive cell 3 times along a , 3 times along b , and 4 times along c . Since the primitive cell contains 2 Ti and 4 O atoms, the supercell contains

$$72 \text{ Ti} + 144 \text{ O} = 216 \text{ atoms.}$$

A neutral DFT+U self-consistent calculation was performed using $U_{\text{eff}} = 3.9$ eV on Ti d states. The total number of electrons from OUTCAR was

$$\text{NELECT} = 1152.$$

This is consistent with the neutral supercell:

$$72 \times 4 + 144 \times 6 = 288 + 864 = 1152.$$

The eigenvalues from EIGENVAL show a highest occupied state at -0.303552 eV and a lowest unoccupied state at 1.931475 eV relative to the Fermi level. This gives a neutral supercell gap of

$$E_g = 2.235027 \text{ eV}.$$

The DOS plot also shows a clear gap, confirming the neutral semiconducting reference before adding the excess electron.

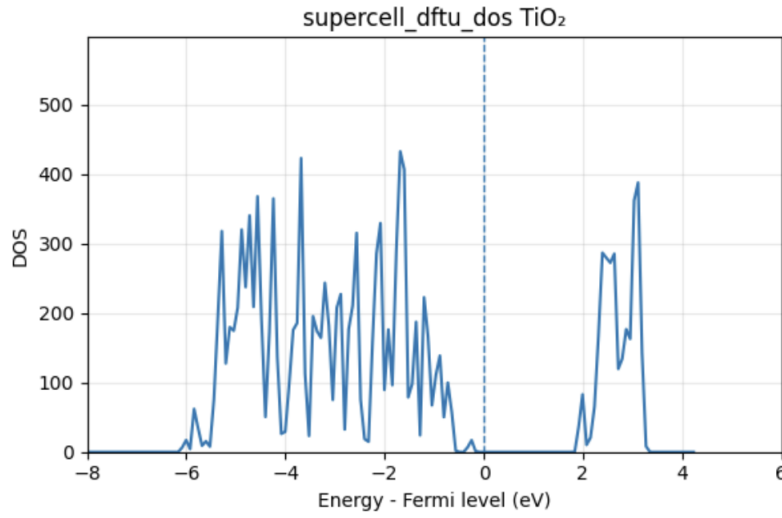


Figure 5: DFT+U DOS of the neutral $3 \times 3 \times 4$ rutile TiO_2 supercell.

6 Polaron trapping A: extra electron using NELECT

For method A, one extra electron was added by manually increasing NELECT from 1152 to 1153. The POSCAR was further modified using a Python script. One Ti atom near the center of the supercell was moved to the beginning of the atom list and treated as a separate Ti species, giving the POSCAR order

Ti Ti O

with counts

1 71 144.

The script also identified the six nearest O atoms around this special Ti atom and displaced them outward by $0.30 \text{ \AA}$. This breaks the local symmetry and helps trap the excess electron on the selected Ti site.

6.1 Large U trapping step

In the large U trapping calculation, $\text{NELECT} = 1153$, confirming that one extra electron was added to the neutral supercell. The eigenvalues show an occupied spin-up state near the gap with occupation 1.0, while the corresponding spin-down state is unoccupied.

The final magnetic moment table shows that only the special Ti atom has a significant local moment. Ion 1 has a total local magnetic moment of

$$1.054 \mu_B,$$

while the other Ti atoms have nearly zero moment. This confirms that the excess electron was trapped mainly on the special Ti site.

6.2 Standard U relaxation

After replacing the large U on the special Ti by the standard value $U_{\text{eff}} = 3.9$ eV, the structure was relaxed again. The eigenvalues show an occupied in-gap state at

$$-0.231823 \text{ eV}$$

relative to the Fermi level with occupation 1.0. The first unoccupied state appears at

$$0.653218 \text{ eV}.$$

The local magnetic moment table shows that only the special Ti ion has a significant magnetic moment, with ion 1 having

$$0.8780 \mu_B.$$

The other Ti atoms are nearly non-magnetic. This confirms that the excess electron remains localized on the special Ti site after returning to standard U. The final DOS also shows the localized in-gap state, consistent with polaron formation.

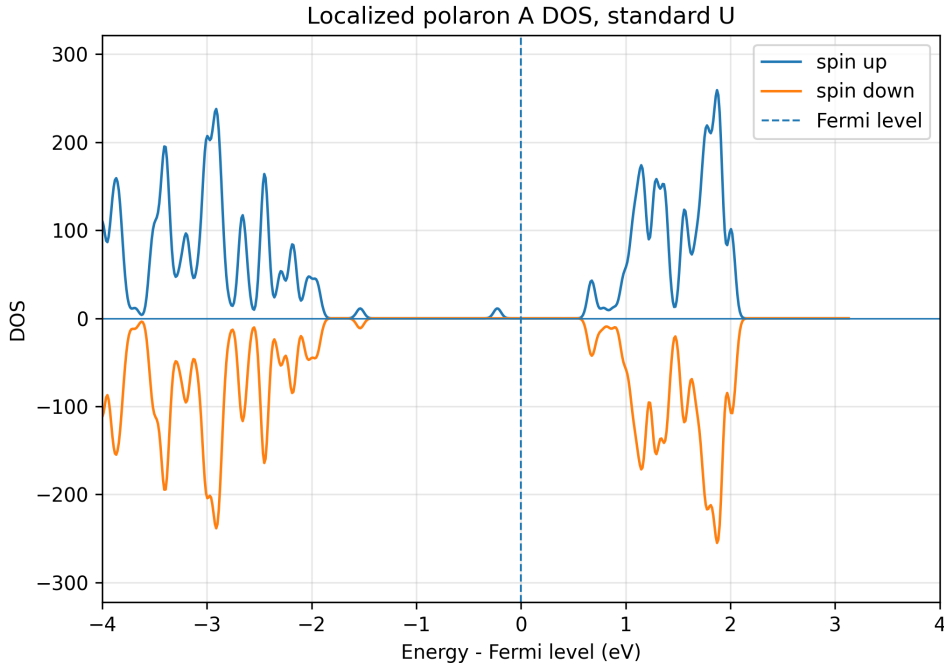


Figure 6: Localized polaron A DOS with standard U.

7 Polaron trapping B: Nb substitution

For method B, one Ti atom in the $3 \times 3 \times 4$ supercell was replaced by Nb. This introduces one extra electron without manually changing `NELECT`. The `POSCAR` and `POTCAR` were modified to include Nb as a separate species, with the `POTCAR` order matching the `POSCAR` species order.

The same trapping procedure as in method A was then repeated. After the standard U relaxation, the eigenvalues and local magnetic moments were inspected.

The final standard U relaxation shows an occupied in-gap state at

$$-0.232871 \text{ eV}$$

relative to the Fermi level with occupation 1.0. The first unoccupied state appears at

$$0.606084 \text{ eV}.$$

This gives a gap of

$$0.838955 \text{ eV}$$

from the occupied polaron state to the first unoccupied state.

The local magnetic moment table shows that only one Ti site has a significant moment. Ion 2 has

$$0.8970 \mu_B,$$

while the other ions have nearly zero moment. This confirms that the excess electron is localized on a Ti site near the Nb substitution. The final DOS also shows the localized in-gap state.

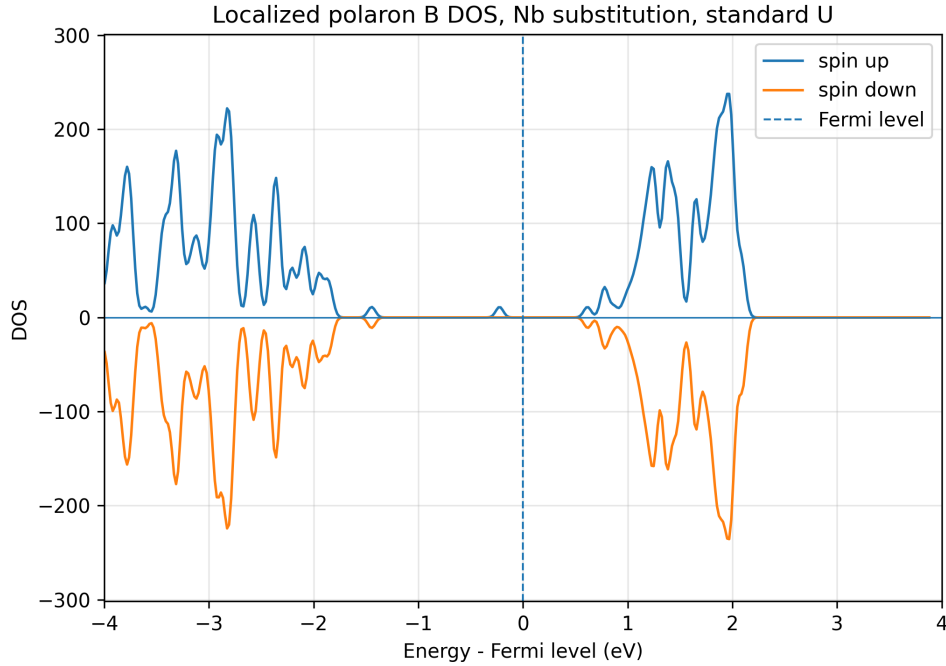


Figure 7: Localized polaron B DOS using Nb substitution with standard U.

For the delocalized B reference, a non-spin-polarized calculation was used with `ISPIN` = 1. The total number of electrons was

$$\text{NELECT} = 1159,$$

and the delocalized reference energy was

$$E_{\text{delocalized}}^B = -1660.77726709 \text{ eV}.$$

8 Polaron formation energy

The polaron formation energy was calculated by comparing the localized polaron solution with the delocalized reference solution:

$$E_{\text{form}} = E_{\text{localized}} - E_{\text{delocalized}}.$$

For the delocalized reference, `ISPIN = 1` was used so that the extra electron remained delocalized in the conduction band.

Table 2: Polaron formation energies for methods A and B.

Method	Localized energy (eV)	Delocalized energy (eV)	Formation energy (eV)
A: NELECT	-1651.995597	-1651.878688	-0.116909
B: Nb substitution	-1660.852487	-1660.777267	-0.075220

Both formation energies are negative. Therefore, the localized polaron solution is more stable than the delocalized solution. In method A, the localized polaron is stable by about

$$0.117 \text{ eV},$$

while in method B it is stable by about

$$0.075 \text{ eV}.$$

9 Conclusion

The PBE bulk calculation shows that rutile TiO_2 is semiconducting, but with an underestimated band gap. Applying DFT+U with $U_{\text{eff}} = 3.9 \text{ eV}$ increases the gap and improves the treatment of Ti d states.

A neutral $3 \times 3 \times 4$ DFT+U supercell was then used as the reference system for polaron calculations. In method A, one extra electron was added using NELECT. In method B, one Ti atom was replaced by Nb. In both cases, the eigenvalues, local magnetic moments, and DOS confirm the formation of a localized small electron polaron.

The calculated formation energies are negative for both methods, showing that the localized polaron is more stable than the delocalized electron. The stabilization is larger for the NELECT method than for the Nb substitution method.